

Soru1

- 1.7. Show that the product of two even signals or of two odd signals is an even signal and that the product of an even and an odd signal is an odd signal.

Let $x(t) = x_1(t)x_2(t)$. If $x_1(t)$ and $x_2(t)$ are both even, then

$$x(-t) = x_1(-t)x_2(-t) = x_1(t)x_2(t) = x(t)$$

and $x(t)$ is even. If $x_1(t)$ and $x_2(t)$ are both odd, then

$$x(-t) = x_1(-t)x_2(-t) = -x_1(t)[-x_2(t)] = x_1(t)x_2(t) = x(t)$$

and $x(t)$ is even. If $x_1(t)$ is even and $x_2(t)$ is odd, then

$$x(-t) = x_1(-t)x_2(-t) = x_1(t)[-x_2(t)] = -x_1(t)x_2(t) = -x(t)$$

and $x(t)$ is odd. Note that in the above proof, variable t represents either a continuous or a discrete variable.

Soru 2 a

- 2.4. The input $x(t)$ and the impulse response $h(t)$ of a continuous time LTI system are given by

$$x(t) = u(t) \quad h(t) = e^{-\alpha t}u(t), \alpha > 0$$

- (a) Compute the output $y(t)$ by Eq. (2.6).
 (b) Compute the output $y(t)$ by Eq. (2.10).
 (a) By Eq. (2.6)

$$y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau) d\tau$$

Functions $x(\tau)$ and $h(t-\tau)$ are shown in Fig. 2-4(a) for $t < 0$ and $t > 0$. From Fig. 2-4(a) we see that for $t < 0$, $x(\tau)$ and $h(t-\tau)$ do not overlap, while for $t > 0$, they overlap from $\tau = 0$ to $\tau = t$. Hence, for $t < 0$, $y(t) = 0$. For $t > 0$, we have

$$\begin{aligned} y(t) &= \int_0^t e^{-\alpha(t-\tau)} d\tau = e^{-\alpha t} \int_0^t e^{\alpha\tau} d\tau \\ &= e^{-\alpha t} \frac{1}{\alpha} (e^{\alpha t} - 1) = \frac{1}{\alpha} (1 - e^{-\alpha t}) \end{aligned}$$

Thus, we can write the output $y(t)$ as

$$y(t) = \frac{1}{\alpha} (1 - e^{-\alpha t})u(t) \quad (2.64)$$

Soru 2 b

(b) By Eq. (2.10)

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

Functions $h(\tau)$ and $x(t - \tau)$ are shown in Fig. 2-4(b) for $t < 0$ and $t > 0$. Again from Fig. 2-4(b) we see that for $t < 0$, $h(\tau)$ and $x(t - \tau)$ do not overlap, while for $t > 0$, they overlap from $\tau = 0$ to $\tau = t$. Hence, for $t < 0$, $y(t) = 0$. For $t > 0$, we have

$$y(t) = \int_0^t e^{-\alpha\tau} d\tau = \frac{1}{\alpha} (1 - e^{-\alpha t})$$

Thus, we can write the output $y(t)$ as

$$y(t) = \frac{1}{\alpha} (1 - e^{-\alpha t}) u(t) \quad (2.65)$$

which is the same as Eq. (2.64).

Soru 3

- 2.12.** Let $y(t)$ be the output of a continuous-time LTI system with input $x(t)$. Find the output of the system if the input is $x'(t)$, where $x'(t)$ is the first derivative of $x(t)$.

From Eq. (2.10)

$$y(t) = h(t) * x(t) = \int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau$$

Differentiating both sides of the above convolution integral with respect to t , we obtain

$$\begin{aligned} y'(t) &= \frac{d}{dt} \left[\int_{-\infty}^{\infty} h(\tau) x(t - \tau) d\tau \right] = \int_{-\infty}^{\infty} \frac{d}{dt} [h(\tau) x(t - \tau) d\tau] \\ &= \int_{-\infty}^{\infty} h(\tau) x'(t - \tau) d\tau = h(t) * x'(t) \end{aligned} \quad (2.76)$$

which indicates that $y'(t)$ is the output of the system when the input is $x'(t)$.